

Federal Reserve Press Conference Language as a Predictor of Short Term Market Movement

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Abstract

This study examines the relationship between the linguistic characteristics of Federal Reserve Chair press conferences and short term S&P 500 behavior. Using 89 transcripts scraped from the Federal Reserve website spanning 2011 to 2026 and corresponding market data from Yahoo Finance, the study analyzed whether language complexity, cautious to definitive language ratios, and tone shifts between prepared remarks and Q&A sessions could predict same day market movement and next day reversals. Language complexity was measured using the Flesch-Kincaid readability score, average sentence length, and jargon frequency, while sentiment analysis was conducted on the sentences using VADER. Multiple regression, logistic regression, and two sample t-tests were used to test four hypotheses. Results showed that linguistic features alone were not significant predictors of market movement. However, when market volatility (VIX) was introduced as a variable, VIX and Flesch-Kincaid grade level were both significant predictors of same day market movement, though the interaction between them was marginally significant ($p = 0.055$), suggesting that language complexity matters more in already volatile market conditions.

1 Introduction

The Federal Reserve Chair holds a press conference after each Federal Open Market Committee (FOMC) meeting to communicate monetary policy decisions to the public. These press conferences are closely watched by investors, analysts, and traders who attempt to interpret not just the policy decision itself but the language used to describe the economic outlook. This study investigates whether measurable linguistic features of these press conferences, such as readability, hedging language, and tone consistency, have a statistically significant relationship with short term S&P 500 movement. If there is a measurable link between language complexity and market volatility, it would suggest that the Fed's choice of words during uncertain periods can carry financial consequences and that clearer communication could reduce the amount of market swings.

2 Related Work

The relationship between central bank communication and market behavior has been studied extensively since the Federal Reserve began increasing transparency in the mid 1990s. Research on FOMC written statements has shown that even small wording changes between consecutive statements can move Treasury yields and equity prices within minutes of release. More recently, NLP tools have been applied to the communication from the Federal Reserve to quantify tone and sentiment at scale, with studies finding that more hawkish or dovish language in statements correlates with directional moves in interest rate futures. However, most of this work focuses on the written statement itself rather than the press conference that follows it. The press conference is a different form of communication because it includes both a scripted opening and an unscripted Q&A where the Chair responds to reporters in real time. This study addresses that gap by analyzing the transcript

as a whole and separately measuring the prepared remarks and Q&A to capture whether tone inconsistency between the two sections has its own effect on markets. Additionally, rather than just measuring directional sentiment, this study focuses on language complexity and readability as predictors, testing if how clearly the Chair communicates is significant compared to what they actually say.

3 Preparation of Study

Data for this study was collected from two sources. Press conference transcripts were scraped from the Federal Reserve website using Python with the requests and pdfplumber libraries. A total of 89 transcripts were successfully downloaded and processed, covering post meeting press conferences from April 2011 through March 2026. Each transcript was split into prepared remarks and Q&A sections using pattern matching on transition phrases such as "I'd be happy to take your questions" and "I'll now take your questions." Market data, including daily S&P 500 opening and closing prices and VIX closing values, are pulled from Yahoo Finance using the yfinance library. For each press conference, the same day return, absolute same day return, next day return, and whether a next day reversal occurred were calculated. Linguistic features were computed for each transcript, including Flesch-Kincaid grade level, average sentence length, jargon frequency per 1,000 words, cautious to definitive language ratio, and sentence level sentiment scores for both the prepared remarks and Q&A sections using VADER. The tone shift was defined as the difference in average sentiment between the two sections. The final merged dataset contained 89 observations with 18 variables.

4 Experimental Hypotheses

This study tests four hypotheses about the relationship between press conference language and market behavior.

Main Hypothesis — Null: The linguistic complexity of the Federal Reserve Chair's press conference has no significant effect on the magnitude of same day S&P 500 movement. Alternative: Higher linguistic complexity is associated with greater same day S&P 500 movement.

Supporting Hypothesis 1 — Null: The cautious to definitive language ratio has no significant effect on the probability of a next day market reversal. Alternative: A higher cautious to definitive ratio is associated with a greater probability of a next day reversal.

Supporting Hypothesis 2 — Null: There is no significant difference in same day S&P 500 movement between press conferences with high tone shifts and those with low tone shifts. Alternative: Larger tone shifts are associated with greater same day market movement.

Supporting Hypothesis 3 — Null: Market volatility (VIX) does not moderate the relationship between language complexity and same day S&P 500 movement. Alternative: The effect of language complexity on market movement is stronger when

VIX is elevated.

5 Study Details

The dataset spans three Federal Reserve Chairs: Ben Bernanke (11 press conferences), Janet Yellen (16), and Jerome Powell (62). The dependent variable for the main hypothesis and supporting hypotheses 2 and 3 was the absolute same day S&P 500 return, which ranged from 0.04% to 3.01%. For supporting hypothesis 1, the dependent variable was a binary reversal indicator, with 39 of 89 press conferences (43.8%) resulting in a next day reversal. All predictor variables were continuous and numeric, including Flesch-Kincaid grade level (range: 7.96 to 13.78), cautious to definitive ratio (0.49 to 2.44), tone shift (-0.09 to 0.21), and VIX close (9.78 to 37.21).

6 Results

A multiple regression using Flesch-Kincaid readability ($p = 0.437$), average sentence length ($p = 0.295$), and jargon frequency ($p = 0.798$) as predictors of same day absolute market movement found no significant results, so this fails to reject the null hypothesis. A logistic regression found that the cautious to definitive language ratio did not significantly predict next day market reversals ($p = 0.502$), and a two sample t-test showed no significant difference in market movement between high and low tone shift press conferences ($p = 0.644$), so this study fails to reject both null hypotheses. However, when market volatility was introduced as a covariate, VIX ($p = 0.009$) and Flesch-Kincaid grade level ($p = 0.048$) were both significant predictors of same day market movement. The interaction term itself was marginally significant ($p = 0.055$), so this study fails to reject the null hypothesis for SH3 at the $\alpha = 0.05$ level, though the effect was in the predicted direction. These results suggest that Federal Reserve language complexity does not independently drive market reactions, but its effect becomes significant in an already volatile market environment where investors are more sensitive to any sort of unclear communication.

7 Discussion

Three of the four hypotheses failed to produce significant results, which makes sense given the number of factors that influence market movement on any given day. The interest rate decision itself, macroeconomic data releases, and major global events all contribute to the same day price action, and these variables were not controlled for in the first three models. This means the language features measured in this study were competing against a large amount of noise in the dependent variable, making it difficult to isolate their individual effects. However, the fourth model gave a significant result that offers some helpful insight. When VIX was included as an additional predictor alongside the Flesch-Kincaid readability score, and the two were tested for interaction, the overall model became statistically significant and explained 35% of the variation in same day S&P 500 movement. This suggests that language complexity does not move markets in isolation, but its effect becomes significant when the market is already in a heightened state of uncertainty. When the VIX is low and markets are calm, investors have time to process a complex or unclear press conference without feeling the need to overreact and panic sell. When the VIX is high and everyone is already on edge, that level of complexity leads to a larger market reaction because investors

who are already under stress are less patient with unclear messaging. The main limitation of this study was the use of the daily open to close returns rather than intraday price data. Because the press conference begins at 2:30 PM and the market closes at 4:00 PM, other events during the trading day could affect the return and add noise to the data.

8 Future Work

Future research could improve this study in a few different ways. The most impactful change would be using intraday price data to measure the S&P 500's movement specifically during the press conference window rather than the full trading day, which would reduce the noise from other market moving events that could also be occurring that day. Another improvement would be adding the interest rate decision and whether it was expected by the market as control variables, since a surprise rate hike will move markets regardless of how the Chair speaks and that effect needs to be separated from the language effect. The linguistic analysis could also be expanded by tracking how specific words or phrases change between consecutive press conferences, which would measure not just the complexity of the language but whether new or unexpected language is what actually drives the reaction. Finally, this study only looked at the S&P 500, but Fed communication affects other parts of the financial system as well. Testing the same hypotheses against Treasury yields or the dollar index could show whether language complexity has a stronger effect on certain assets compared to others.

9 Conclusion

This study set out to determine whether the language used by the Federal Reserve Chair during press conferences has a measurable effect on short term S&P 500 movement. After scraping 89 transcripts from the Fed's website and computing linguistic features such as readability, hedging language, and tone shifts, the results showed that none of these features were significant predictors of market movement on their own. However, when market volatility was factored in, the model became significant, with both VIX and Flesch-Kincaid grade level producing p values below 0.05. This means that how the Federal Reserve Chair speaks does not matter much when markets are calm, but it starts to matter when markets are volatile and investors are worried. This has practical implications for how the Federal Reserve approaches communication during periods of high volatility, and it suggests that prioritizing clarity during those moments could help prevent unnecessary market shifts.

References

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